

level will store more data than the one with a higher level.

The error-correction levels can be chosen as per the usage environment. For instance, if the QR code is to be used in a factory to track batches, more error correction (level H) is required. On the other hand, in a clean environment such as a print magazine, level L can be used.

QR codes for marketing

QR codes provide a convenient one-step process for directing users to a website, phone number, promotions or other information. Besides, these are environment-friendly as they make an efficient use of the printed material. Creating them costs virtually nothing when used in a number of versatile marketing and advertising instruments such as print ads, brochures, visiting cards and event displays.

Measuring a marketing campaign.

A marketing campaign becomes redundant if there is no tool to track its effectiveness. Using QR codes, it is easy to track and measure marketing campaigns with the help of Web analytic tools.

If used effectively, the advantages of QR code can be harnessed by any organisation that caters to a smartphone-using target audience—be it a retail business, a non-profit organisation, a membership association or an educational institution.

Sharing. A QR code can be used to share a video or a landing page. It contains enough alphanumeric characters to link to a website. So it can display the download link of an app, which, in turn, can promote your business.

Community. Applications can be used to create QR code that links to a social networking functionality such as 'Like' button of a community page on Facebook.

QR codes can be used in visiting cards and on products to include information such as price and manufacturing details. These can contain a lot of information such as the location of your business and the category of products you offer. They can also be used in an innovative way for linking to application installation links, customer feedback forms, locations on

Innovative uses

Emudhra is launching a mobile wallet service called YpayCash in which the payment application scans the QR code on products and then the transaction is performed.

HP has recently partnered with Mindfire to provide Smartstream services to HP users. Mindfire will help HP include QR code in direct e-mails sent to users. A unique landing page will be personalised for users. Using unique QR codes, users will learn about new HP products. They will also have the facility to create and print QR codes directly into their e-mails with the help of HP software tools.

Cottonworld, a leading apparel brand, offered discounts to its customers on the occasion of Diwali last year by giving away cards with QR codes printed on them. Users could scan the codes with their smartphones and avail discounts.

Google Maps, etc. You can track your customers using QR codes and find out where the purchase was made.

Major airlines have been using QR code in their air tickets. Also, while registering with Google Places, you get a QR code linking to the website.

Disadvantages of QR codes

Any new technology has its own set of problems that calls for attention and subsequent solutions. In the case of QR codes, it has been argued that they may not work due to the very nature of the technology. The very purpose of QR codes is to encrypt data within the code, which provides scope for cyber criminals to cloak smartphone attacks. There is no way to know whether the code is genuine and links to a legitimate website or an infected one, a malicious app or a phishing site.

A type of infected QR code downloads an app that introduces a hidden texting charge in your monthly cellphone bill. QR codes can also be used to gain full access to a smartphone—including Internet access, camera, GPS, read/write local storage and contact data. All the data on a smartphone can be stolen by remotely transferring it, putting the user at the risk of identity theft.

QR code in India

A Google Trends report on search patterns for QR codes puts India at the eighth place in the world. In India, one of the first companies to use QR code in advertising was Ford, which used it to promote Ford Figo. Users were asked to download an application that read QR code by sending a text message. They could then play a video of the Figo on their mobile device.

Error-correction Levels of QR Code

Level	Percentage
Level L	Approx. 7%
Level M	Approx. 15%
Level Q	Approx. 25%
Level H	Approx. 30%

Marketing companies such as Customer Centria have used QR codes in promotional campaigns of Godrej's social networking site Gojiyo and Taj Hotel's holiday packages.

Some Indian portals such as www.flick2know.com provide free generation and implementation of QR code.

The future

QR codes are a wonderful marketing tool if used effectively. These should be carefully and cleverly placed and used to the advantage of the marketer. A QR code is not effective if it doesn't lead to mobile-friendly content. Placing QR codes on billboards where traffic moves fast makes it impossible to stop and scan. Their use in subway platforms where there is no signal also makes them redundant. When used in fliers, banners, etc in an event, it is essential to have WiFi or other Internet access at the venue.

As of now, QR codes require connection to the Internet to give information. We await the arrival of QR codes that can store much more data than the present generation and don't require an Internet connection to furnish that. ●

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